

WEEK-1 : REQUIREMENT ANALYSIS

Understanding E-Commerce Business :

An online shopping system is used to connect customers with products and sellers through a digital platform. Customers can create an account, look for products, place orders and track their purchases.

For developing the database, we need to identify what information should be stored about the customers. The customer module mainly contains personal and contact details. These details can later be connected with orders, payments and product reviews.

The Process Flow:

- **Browse / Search Product:**

The customer searches for the required product and checks the available products on the e-commerce platform.

- **Add to Cart:**

After selecting a suitable product, the customer adds it to the shopping cart for purchase.

- **Order Placed:**

The customer reviews the selected items, provides the delivery address, and confirms the order.

- **Checkout / Payment:**

The order moves to checkout, where the customer selects a payment method and completes the payment.

- **Delivery:**

Once the payment is completed, the order is processed and the selected product is sent for delivery to the customer's address.

- **Review & Support:**

After receiving the product, the customer can give feedback or contact support for issues such as returns, replacements, or product-related assistance.

Entities :

ENTITY	DESCRIPTION	KEY ATTRIBUTES	KEY RELATIONSHIP
Customer	Details of users who purchase products through the platform.	customer_id (PK), customer_name, email, phone, address, registration_date	One-to-Many with Order, Review
Seller	Details of sellers who make products available on the platform.	seller_id (PK), company_name, email, phone, rating	One-to-Many with Product
Product	Information about products listed for customers.	product_id (PK), seller_id (FK), product_name, price, stock_quantity, category	Many-to-One with Seller
Order	Records the purchases made by customers.	order_id (PK), customer_id (FK), order_date, status, total_amount, shipping_address	Many-to-One with Customer
Payment	Details of payments associated with customer order	payment_id (PK), order_id (FK), payment_method, status, amount, payment_date	One-to-One with Order
Review	Feedback and ratings given by customers for products	review_id (PK), customer_id (FK), product_id (FK), rating, comment, created_at	Many-to-One with Customer and Product

Customer Table Creation:

```
SQL> CONNECT C##ECOMMERCE/3012
```

```
Connected.
```

```
SQL> SHOW USER;
```

```
USER is "C##ECOMMERCE"
```

```
SQL> desc customer;
```

Name	Null?	Type
CUSTOMER_ID	NOT NULL	NUMBER(38)
CUSTOMER_NAME	NOT NULL	VARCHAR2(50)
EMAIL	NOT NULL	VARCHAR2(100)
PHONE	NOT NULL	VARCHAR2(15)
ADDRESS	NOT NULL	VARCHAR2(200)
REGISTRATION_DATE	NOT NULL	DATE

Insert Values:

```
SQL> INSERT INTO customer(customer_id,customer_name,email,phone,address,registration_date)VALUES(&customer_id,&'customer_name',&'email',&phone,&'address',DATE '&registration_date');
```

```
Enter value for customer_id: 1
```

```
Enter value for customer_name: Arun Kumar
```

```
Enter value for email: arun@gmail.com
```

```
Enter value for phone: 9876543210
```

```
Enter value for address: Chennai
```

```
Enter value for registration_date: 2025-01-10
```

```
old 1: INSERT INTO customer(customer_id,customer_name,email,phone,address,registration_date)VALUES(&customer_id,&'customer_name',&'email',&phone,&'address',DATE '&registration_date')
```

```
new 1: INSERT INTO customer(customer_id,customer_name,email,phone,address,registration_date)VALUES(1,'Arun Kumar','arun@gmail.com',9876543210,'Chennai',DATE '2025-01-10')
```

```
1 row created.
```

Customer Table:

```
SQL> SELECT * FROM CUSTOMER;
```

CUSTOMER_ID	CUSTOMER_NAME	EMAIL	PHONE	ADDRESS	REGISTRATION
1	Arun Kumar	arun@gmail.com	9876543210	Chennai	10-JAN-25
2	Priya Sharma	priya@gmail.com	9876543211	Bangalore	15-JAN-25
3	Rahul Raj	rahul@gmail.com	9876543212	Coimbatore	05-FEB-25
4	Karthik M	karthik@gmail.com	9876543214	Chennai	20-FEB-25
5	Anjali Devi	anjali@gmail.com	9876543215	Salem	01-MAR-25
6	Vijay Kumar	vijay@gmail.com	9876543216	Chennai	08-MAR-25
7	Meeena K	meena@gmail.com	9876543219	Vellore	02-APR-25
8	Ajay Kumar	ajay@gmail.com	9876543220	Chennai	10-APR-25
9	Manoj R	manoj@gmail.com	9876543222	Tirunelveli	01-MAY-25
10	Nithya S	nithya@gmail.com	9756864237	Coimbatore	12-MAY-25
11	Ravi Chandran	ravi@gmail.com	47589364470	Pondicherry	20-MAY-25
12	Suresh Babu	suresh@gmail.com	7895634786	Erode	25-JUN-25
13	Akash Kumar	akash@gmail.com	9078237684	Chennai	16-JUL-25
14	Gayathri R	gayu@gmail.com	8964785690	Salem	17-AUG-25
15	Ishitha R	ishitha@gmail.com	8976486397	Vellore	10-SEP-25

```
15 rows selected.
```

Update Method:

```
SQL> UPDATE customer SET phone = 9085682356 WHERE customer_id = 15;  
1 row updated.
```

Conclusion

In this week, the basic requirements of the e-commerce database were identified and the customer module was developed. The customer table was created to store important details such as customer ID, name, email, phone number, address, and registration date. SQL commands were used to insert and update customer information. This provides a basic database structure for managing customer details and can be extended with other e-commerce modules in the upcoming weeks.